

ForeDance: Structured Streaming Dance Generation with Anticipatory Music Conditioning

Abstract—Streaming dance must prepare for upcoming music from audio heard so far while continuing movements that have already been committed and can no longer be revised. We present ForeDance, a framework that generates robot-native dance from music as it arrives. A frozen music foundation model forecasts future musical context using heard audio alone. A kinematics-guided structure–detail representation gives whole-body organization and local articulation complementary roles, while Commit Forcing trains continuation from jointly generated motion and its resulting reference state. Together, these components couple musical anticipation with successive motion decisions. On 60-second FineDance generations, a frozen music–motion evaluator favors the intended soundtrack over alternative tracks in approximately 75% of pairwise comparisons. Separate matched training comparisons identify a tradeoff between motion distribution fidelity and smoothness. These findings support soundtrack correspondence in continuously generated references within a framework that connects musical foresight to committed movement.

I. INTRODUCTION

Dance preparation begins before a visible accent: weight shifts precede a step, and the body transitions toward a pose before reaching it. A system dancing to music as it arrives must choose these movements before hearing all the sound they will accompany. Waiting for an event leaves less time to prepare; reading the future recording changes the task into offline generation. Streaming further requires each new movement to continue from emitted motion that can no longer be revised. Musical anticipation and motion continuation are therefore coupled requirements of streaming dance generation.

Music-conditioned choreography has advanced through expressive motion models and large paired datasets [1]–[4]. Streaming systems such as DiscoForcing [5] encode incoming audio causally, supporting motion generation as the soundtrack unfolds. Our question is how to supply prospective musical context when future audio is unavailable. We investigate this question in robot-native dance: the model produces reference motion in the humanoid’s own configuration space, so musical choices must be expressed through its body and continued across successive planning decisions.

We present ForeDance, a framework that conditions streaming dance on predicted future music states. A frozen Magenta RealTime 2 (MRT-2) music foundation model [6] rolls forward from the audio received so far. This transfers a general musical continuation prior into dance generation: paired dance data teaches the motion model to use the prior, while musical prediction is learned through large-scale music pretraining. To our knowledge, ForeDance is the first streaming dance generation framework to explicitly condition

motion on future music states forecast by a pretrained music foundation model using past and present audio alone. The predicted states guide both whole-body organization and complementary articulation.

Realizing this design requires a representation that organizes movement and a training procedure that accounts for earlier generated choices. Our kinematics-guided structure–detail representation couples discrete structure and continuous detail with complementary reconstruction objectives. Structure-only reconstruction preserves designated body organization and endpoint placement; their combination reconstructs full motion through a shared decoder. Building on hybrid motion representations [7], this supervision specifies what the two components should preserve. Commit Forcing then trains continuation after jointly sampling and committing both components. Their decoded movement determines the next starting state, pairing generated history with the reference state it actually induces.

At each decision, ForeDance plans a short segment, commits its first segment, and updates the context for the next plan (Fig. 1). A reference interface delivers the resulting stream to a fixed SONIC whole-body controller [8]. The generator continues from committed references; measured robot feedback is handled by the controller. This keeps the motion-generation mechanism explicit while providing a path to motion tracking.

Our contributions are:

- **Anticipatory music conditioning for streaming dance.** Predicted future states from a frozen music foundation model provide musical lookahead using arrived audio alone.
- **Structure–detail representation and paired training.** Complementary reconstruction objectives establish structural and detail roles for coupled music-conditioned generation through a shared decoder.
- **Commit Forcing for streaming continuation.** Jointly generated structure and detail define both the committed history and its resulting reference state for continuation training.

Evaluations of 60s FineDance generations find that a frozen music–motion evaluator favors the intended soundtrack in approximately 75% of pairwise comparisons. A separate, complete 18-track study characterizes the effects of history regularization on motion fidelity, diversity, and smoothness. Together with rendered references, these studies examine both musical correspondence and the resulting motion. Section V distinguishes these findings from the

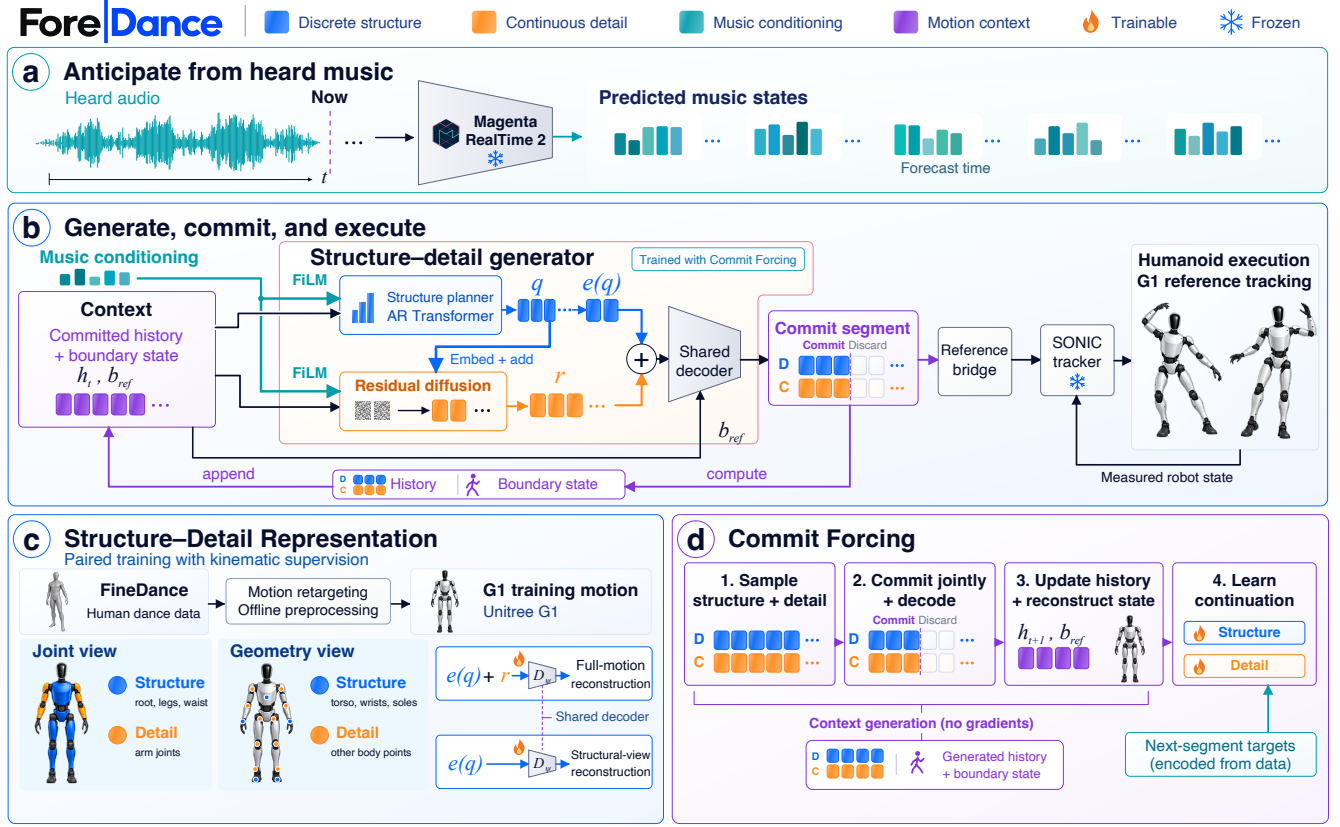


Fig. 1. **ForeDance overview.** **a:** Heard audio supplies a causal forecast of upcoming music states. **b:** The music condition enters both generation branches through FiLM; sampled structure also conditions detail. A shared decoder maps their sum to reference motion. Joint commitment updates code history and the reference-derived boundary state, while measured robot feedback stays within the fixed tracking loop. **c:** Paired reconstruction gives structure and detail complementary kinematic roles. **d:** Commit Forcing trains continuation from jointly generated history and its resulting state. Robot poses and time ranges are schematic, not experimental or hardware execution evidence.

comparisons needed to attribute forecast and component benefits.

II. RELATED WORK

A. Observed and Predicted Music Conditions

Music-to-dance models combine acoustic descriptors [9] or learned features [10], [11] with choreography models. FACT and Bailando model coordinated sequences and choreographic memory [1], [12]; EDGE and LODGE develop diffusion-based and long-form generation [2], [13]. FineDance, MEGADance, and SoulNet address fine-grained motion, genre, and music–motion alignment [3], [4], [14].

DiscoForcing [5] uses a task-specific causal VQ-PAE encoder to describe heard audio through rhythmic tokens and periodic alignment. We instead condition on explicit future states from a generative music prior. MRT-2 is pretrained on approximately 71,000 hours of music [6], providing continuation knowledge beyond our paired dance training. Foundation-model features already support dance in EDGE through Jukebox [2], [15] and in DGFM [16]; earlier robot systems also predict beat timing [17]. Our distinction is the use of generative music-state forecasts under a causal audio constraint. Live Music Models [18] and MRT-2 supply the

music-generation foundation; we transfer its predictions to structured streaming motion.

B. Structure–Detail Representation and Training

Discrete tokens support motion generation in T2M-GPT and MoMask [19]–[21]; MLD uses continuous latent diffusion [22], and DisCoRD uses flow-based decoding from discrete tokens [23]. Most closely, DC-Motion [7] combines discrete structure with continuous residuals. We build on this hybrid principle, using designated robot-kinematic supervision to give the components complementary roles through a shared, boundary-conditioned decoder. Neural Categorical Priors and GPC learn vocabularies for physical control [24], [25]; our latents instead decode into explicit motion references for a separate tracker.

C. Continuation from Generated Context

Scheduled Sampling and Professor Forcing address training–sampling mismatch [26], [27]. Diffusion Forcing varies per-token noise [28]; MotionStream and MotionStreamer develop causal motion generation [29], [30]. MotionStreamer’s Two-Forward strategy uses first-pass predictions in second-pass history. Commit Forcing constructs the joint structure–detail history–state transition: sampled components enter history together, and their decoded committed

motion determines the next reference state. Its distinction is this coupled update. Diffusion Policy and Real-Time Chunking address committed actions during plan updates [31], [32]; CoF constructs the context under which continuation is learned.

a) Robot execution: Reusable trackers support expressive motion [8], [33]–[35], while HARMON and RoboPerfom introduce language and audio conditioning [36], [37]. PhysDiff, terrain-aware reference generation, and PARC incorporate physical correction or adaptation [38]–[40]. Our fixed-controller interface keeps reference generation separate from measured-state feedback, allowing their outputs to be examined independently.

III. METHOD

ForeDance couples three decisions: what music to anticipate, how to express the response as structured motion, and how to continue after committing it. Figure 1 shows the complete system; the following figures expand the conditioning, reconstruction, and continuation mechanisms.

a) Streaming task: At decision time τ , $a_{\leq\tau}$ is the heard audio, h is committed code history, and b is the pose and velocity state reconstructed from the last committed reference frames. A frozen music model predicts $m^{\text{pred}} = F(a_{\leq\tau})$, and the generator samples

$$(q_{1:H}, r_{1:H}) \sim p_\theta(q, r \mid h, b, m^{\text{pred}}). \quad (1)$$

Here q is discrete structure and r is continuous detail. Their sum decodes to native G1 motion. Both streams commit the same first $C < H$ steps; the remaining plan is discarded. We use $H = 8$ and $C = 4$ at 15 latent steps per second, yielding 16 planned and eight committed frames at 30 Hz.

A. Anticipatory Music Conditioning

SpectroStream tokenizes heard audio, and frozen MRT-2 Small rolls forward to predict musical states [6]. Its music pretraining supplies the continuation prior; paired dance training learns how to turn that prior into movement. Fourteen forecast states at 25 Hz cover approximately 0.56 s, alongside the 0.53 s motion plan. Future waveform samples are never used as input.

a) Align time before modulating features: Forecast and motion steps have different rates, so matching array indices would assign the wrong musical time. For state u_j with physical lead ℓ_j relative to the decision, we form

$$c_j = W \text{LN}(u_j) + \phi(\ell_j). \quad (2)$$

Here W projects the music features and ϕ encodes their lead time. We linearly interpolate neighboring projected states at motion lead $d_i = i/15$ to obtain $\hat{c}_i$ (Fig. 2). A coverage mask m_i is one only within the valid forecast support and zero when the condition is absent or no longer covers that motion time.

Each conditioned block adds a gated, frame-wise modulation [41]:

$$v'_i = v_i + m_i g \odot [s(\hat{c}_i) \odot \text{LN}(v_i) + t(\hat{c}_i)]. \quad (3)$$

The learned maps s, t predict scale and shift, and g is a learned channel gate. When $m_i = 0$, the block retains its original activation exactly. Separate modulators condition the structural Transformer blocks and residual diffusion heads using the same physical-time alignment. The generator and these mappings are trained; MRT-2 remains frozen.

B. Structure–Detail Representation and Paired Training

An encoder maps native motion and its starting state to $z = E_\phi(x, b)$. Quantization selects $q = Q(z)$ with codebook embedding $e(q)$, and a continuous residual retains the remaining information:

$$z = e(q) + r. \quad (4)$$

Both components are decoded by one boundary-conditioned decoder $D_\psi(\cdot, b)$. Their roles are learned through complementary reconstruction objectives, rather than separate anatomical decoders.

a) Learning complementary roles: We train two reconstructions with the same decoder and starting state:

$$x_f = D_\psi(e(q) + r, b), \quad x_s = D_\psi(e(q), b). \quad (5)$$

The full path preserves complete movement, while the structure-only path gives the discrete anchor an explicit kinematic responsibility (Fig. 3). Codebook and commitment losses train the quantized representation; Appendix I specifies the gradient routes.

The full loss supervises native coordinates, native velocities, forward-kinematics positions, and geometry velocities. Each block is standardized with training-set statistics and receives equal weight. The structural loss uses the corresponding selected coordinates: root, legs, and waist in the native views, and torso, wrists, and soles in the geometry views. Consequently, wrist placement can be structural while arm-joint articulation remains part of the complementary native view. Auxiliary structural reconstruction under residual corruption and a boundary-transition loss reinforce these objectives; their settings and exact partitions remain in Appendix I.

b) Coupled motion generation: A causal Transformer samples q . Its sampled IDs are embedded and added to the residual model’s input features; a diffusion generator [42] samples r using this structure, motion history, boundary state, and aligned music. The codebook embedding and sampled residual then sum at the shared decoder to produce 38-dimensional native motion. Joint commitment preserves the pairing of the two streams across planning steps.

C. Commit Forcing

Teacher Forcing supplies recorded history and a recorded starting state; streaming inference supplies both from the model’s earlier output. Commit Forcing changes this conditioning transition while retaining recorded supervision for the next segment (Fig. 4).

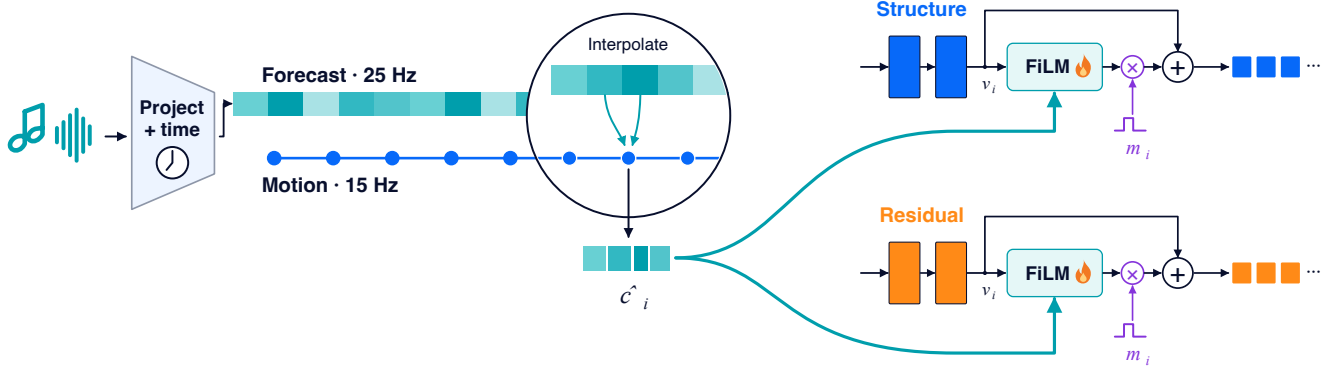


Fig. 2. **Physical-time music conditioning.** Forecast features are projected with their lead times and interpolated at motion-step times. The aligned features drive separate residual FiLM modules in the structure and residual branches. Coverage masking bypasses modulation outside valid support.

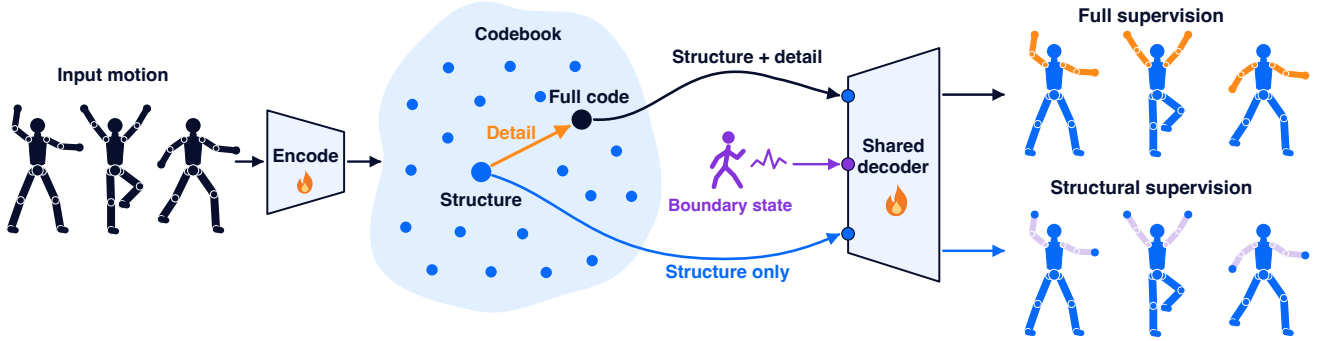


Fig. 3. **Learning complementary structure and detail.** A discrete anchor and continuous offset form the full code. Full and structure-only reconstructions share one boundary-conditioned decoder. Pose colors illustrate complete versus selected supervision; pale limbs remain in the reconstruction but are outside the selected native view. Geometry is schematic; flames mark learned modules.

a) Construct the next history and state together: A first pass samples $\tilde{q}, \tilde{r}$ with the inference samplers under stop-gradient. Their common committed segment updates the two parts of the next context:

$$(\tilde{h}^+, \tilde{b}^+) = U(h, b, \tilde{q}_{1:C}, \tilde{r}_{1:C}). \quad (6)$$

U appends the committed codes to history and reconstructs the boundary state from their decoded reference motion. Context replacement selects this entire pair together, so generated history is accompanied by the state of the same commitment. Each supplied training example constructs one such transition independently; the first pass is not a recursive long rollout.

b) Keep the target, change its residual coordinates: The next recorded structural target q^* and full latent $z^* = e(q^*) + r^*$ remain fixed. The next structure model is supervised against q^* , while its sampled structure $\hat{q}$ conditions residual denoising. We express the residual target relative to that sampled embedding:

$$r_{\text{rebased}}^* = z^* - e(\hat{q}), \quad e(\hat{q}) + r_{\text{rebased}}^* = z^*. \quad (7)$$

Thus the structure and residual losses retain a common recorded target latent despite a changed structural sample. This is latent rebasing, not target re-encoding under $\tilde{b}^+$: the decoder also depends on its boundary state, so latent equality does not imply identical decoded motion.

c) History regularization: After context construction, full-history corruption (FHC) perturbs discrete embeddings and residuals across the motion history. It preserves boundary state, music, masks, and timestamps, and is disabled during inference. Appendix I specifies this separate training regularizer.

D. Robot Reference Integration and Execution

Figure 5 separates reference delivery from controller feedback. The committed native motion provides the reference interface to a separate, pretrained SONIC whole-body controller [8]. A reference bridge reconstructs root motion in the fixed sequence coordinate system, maps joint references into the controller convention, and resamples the 30 Hz motion onto a continuous 50 Hz grid. Appendix I defines the coordinates and state reconstruction. Only committed motion

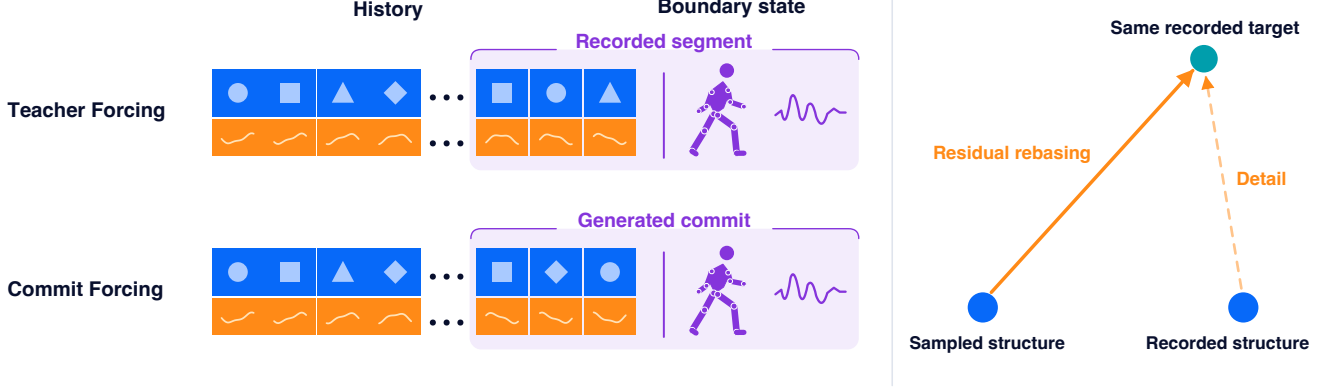


Fig. 4. **Paired context and residual rebasing.** TF takes recent history and boundary state from a recorded segment; CoF derives both from the same generated commitment. At right, residual rebasing moves the residual’s origin to the sampled structure while retaining the recorded target latent. Tile counts and geometric positions are schematic.

is delivered; the uncommitted suffix is discarded before the next plan and is not sent to the controller.

SONIC remains fixed and combines the consumed reference with measured robot state to produce control actions. The generator separately updates its latent history and boundary state from committed reference motion. Appendix IV specifies the delivery and measurement protocol.

IV. EXPERIMENTS

We separate evaluation of the music–motion evaluator on reference motion from its use to assess generated dance. Beat alignment and motion-quality measures complement learned correspondence. Supporting history-training comparisons examine motion quality; the core ablations still awaiting results are marked explicitly in Section IV-D.

A. Setup and Metrics

Evaluator validation uses reference motion from 30 content groups, comprising 3,661 windows and 15 disjoint song pairs per discrimination task. Generator correspondence uses 99 generated 60 s FineDance trajectories: eleven tracks, three independently trained generators, and three sampling seeds per generator. The separate beat-alignment and motion-quality study uses all 18 archived tracks, a common 23 s segment, and the same three-by-three training/sampling repeat structure, giving 162 outputs per configuration. All motion is represented in native G1 space. Appendices I and II specify the training and cohort protocols.

a) Learned music–motion correspondence: Motivated by music–motion retrieval [4], we use a frozen G1-native evaluator trained independently of the generator. Its score $s_w(x, a)$ averages cosine similarities from three evaluator instances for five-second music–motion windows. For intended music a_i and an alternative track a_j , the correspondence margin is

$$\delta_{ijw} = s_w(x_i, a_i) - s_w(x_i, a_j). \quad (8)$$

Pairwise accuracy counts positive margins, with half credit for ties; random ordering gives 50%. The evaluator-validation

TABLE I
EVALUATOR VALIDATION ON REFERENCE MOTION. THE FROZEN EVALUATOR RANKS INTENDED MUSIC AGAINST ORDINARY OR MUSICALLY SIMILAR ALTERNATIVES. PAIRWISE ACCURACY IS IN PERCENT; RANDOM ORDERING IS 50%.

Discrimination task	Accuracy (%)
Ordinary alternative music	68.41
Similar alternative music	64.08

tasks use ordinary or musically similar alternative tracks. Generator scoring compares each fixed motion against its intended music and all ten alternatives in its fixed gallery, preserving native audio offsets. Windows and sampling repeats are aggregated within training-seed/query-track cells. Appendix II-A defines the frozen evaluator.

b) Beat alignment and motion quality: Beat Alignment Score (BAS) measures the proximity of motion beats to music beats. We use local minima of smoothed mean forward-kinematics keypoint speed and a motion-to-music Gaussian matching score with a 0.10 s scale; Appendix II-A.0.e gives the exact definition. BAS measures rhythmic timing, while learned correspondence assesses soundtrack preference. Kinetic and geometric Fréchet distances (FID_k , FID_g) compare generated and reference distributions of motion dynamics and body configurations. Div_k and Div_g measure spread in the same feature spaces. PFC is a contact-related physical-consistency proxy, complemented by root/contact and jerk diagnostics. The 23 s and 60 s cohorts are analyzed separately.

B. Evaluator Validation

Table I evaluates the score itself on reference motion. Pairwise accuracy is 68.41% against ordinary alternative music and 64.08% against musically similar alternatives. Both exceed random ordering in point estimate, with the similar-music task providing a more demanding correspondence check. These results characterize the evaluator’s ability to

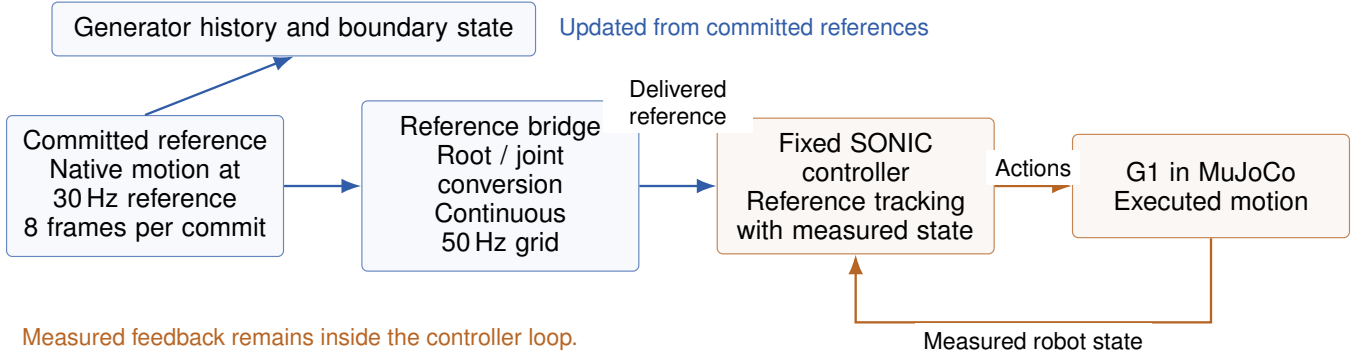


Fig. 5. **From committed references to motion tracking.** The bridge converts the reference stream onto the controller grid. Committed references update the generator context, whereas measured robot state closes the SONIC tracking loop.

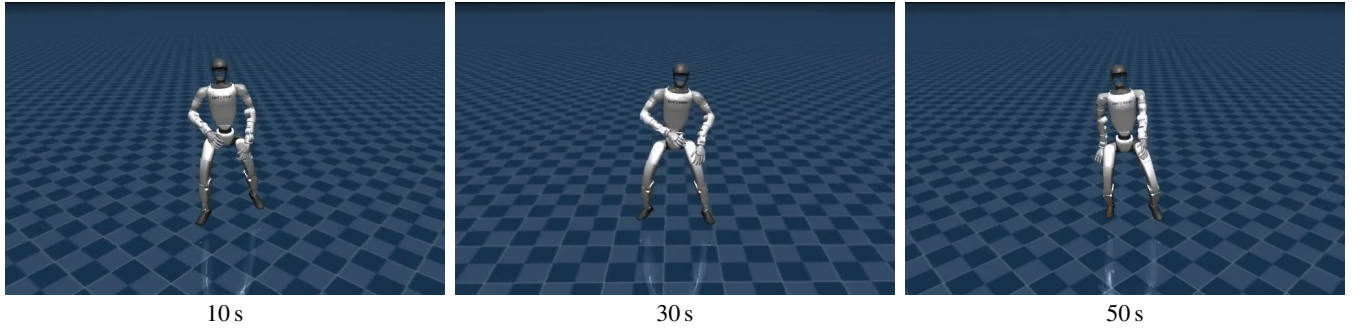


Fig. 6. **A 60-second ForeDance generation.** Original frames at 10, 30, and 50 s from a selected FineDance sequence. The images show rendered reference motion.

TABLE II
MUSIC ALIGNMENT OF FOREDANCE GENERATIONS. LEARNED CORRESPONDENCE IS PAIRWISE SOUNDTRACK ACCURACY; BAS MEASURES MOTION-TO-MUSIC BEAT ALIGNMENT. HIGHER IS FAVORABLE FOR BOTH METRICS; THEIR EVALUATION COHORTS AND SCORE SCALES DIFFER.

Measure	Evaluation cohort	Score
Learned correspondence	11 tracks, 60 s	75.01%
Beat alignment (BAS)	18 tracks, 23 s	0.4419

distinguish soundtracks, rather than the quality of ForeDance generations.

C. Generator Evaluation

Table II reports learned soundtrack correspondence and beat alignment for generated motion. The intended soundtrack wins 75.01% of pairwise comparisons, with a mean correct-minus-alternative score margin of 0.1783. Every generator training seed has a positive mean margin. The separate 18-track study yields BAS 0.4419. These complementary measures characterize soundtrack preference and rhythmic alignment; their different cohorts preclude treating the values as a single pooled evaluation. Appendix III retains the detailed score statistics and descriptive intervals.

a) *History-training comparison:* Table III compares clean history, history dropout, and full-history corruption

(FHC), holding the remaining settings fixed. Relative to clean history, FHC reduces kinetic/geometric FID by 21.9%/15.0%, with both directions favorable in all three training seeds, and increases kinetic diversity. BAS rises from 0.4352 to 0.4419; this is a descriptive change, not a significance claim. PFC and jerk favor clean history, so the distribution gains coexist with a smoothness tradeoff. This comparison examines history regularization, rather than isolating the benefit of musical anticipation or Commit Forcing.

b) *Qualitative generation:* Figure 6 shows reference poses at 10, 30, and 50 s from one selected 60 s rollout. These original frames illustrate body configuration across the sequence; they do not measure temporal smoothness.

D. Pending Ablation Studies

TODO for this working draft: the following controlled studies await results and are not counted as completed evidence.

a) *TODO: musical anticipation:* Compare forecast-conditioned and heard-music-conditioned generators with matched training, tracks, and sampling. Report learned correspondence, BAS, and motion quality to isolate the value of predictive musical context.

b) *TODO: paired representation training:* Complete current-representation reconstruction and intervention results to test whether structure and detail serve their intended complementary roles. Include non-collapse checks and matched

TABLE III

MOTION QUALITY AND BEAT ALIGNMENT UNDER MATCHED HISTORY TRAINING. ALL THREE CONFIGURATIONS SHARE INITIALIZATION, ARCHITECTURE, MUSIC CONDITIONING, THE COMPLETE 18-TRACK COHORT, COMMON 23-SECOND SEGMENTS, THREE TRAINING SEEDS, AND THREE SAMPLING SEEDS. LOWER IS FAVORABLE FOR FID AND PFC; HIGHER BAS INDICATES CLOSER BEAT ALIGNMENT. DIVERSITY DESCRIBES MOTION SPREAD AND IS INTERPRETED TOGETHER WITH FIDELITY.

History training	FID _k	FID _g	Div _k	Div _g	BAS	PFC
Clean history	208.599	0.482	7.495	0.493	0.4352	0.143
History dropout	164.761	0.469	7.801	0.473	0.4338	0.182
History corruption	162.831	0.410	8.626	0.508	0.4419	0.167

supervision comparisons.

c) *TODO: single-representation controls:* Compare independently trained discrete-only, continuous-only, and combined structure–detail generators under matched settings. Removing a decoder branch does not substitute for a separately trained baseline.

d) *TODO: committed continuation:* Compare Commit Forcing with Teacher Forcing on matched 60 s free-running sequences. Report aggregate quality and each of the three non-overlapping 20 s windows, including commitment-boundary behavior and matched qualitative sequences.

E. Supporting Execution Diagnostics

The interface in Section III-D separates the generated, resampled, controller-consumed and measured trajectories. The supporting study replays four saved trajectories from one FineDance track through fixed SONIC in MuJoCo. These trajectories use an earlier 34-dimensional representation, separate from the current 38-dimensional generation cohort; audio inference is not run during replay.

After excluding the first two seconds and aligning joint trajectories, median amplitude retention ranges from 0.858 to 0.901, and median squared-velocity retention from 0.411 to 0.496. Position tracking therefore coexists with dynamic attenuation. These diagnostics characterize the reference-to-execution path; they do not establish current-model hardware performance or comparative musical benefit. Appendix IV retains the complete measurements and protocol.

V. DISCUSSION

a) *Supported findings and attribution:* The generator studies characterize soundtrack correspondence, beat alignment, and a distribution–smoothness tradeoff in history regularization. The evaluator’s reference-motion validation is a separate measurement. Attributing gains to anticipation, representation design, or committed continuation requires the controlled studies listed in Section IV-D. Broader music pretraining supplies a transferable prediction prior, while improved out-of-distribution dance generalization remains a hypothesis. The historical replay diagnostics characterize tracking separately from these current reference-generation results.

b) *Evaluation scope and selection:* The evaluator configuration was selected after inspection of its 30-content-group calibration cohort. For compact reporting, generator music scores use the higher complete initialization result

for each metric after evaluation; all tracks and sampling repeats in the respective cohort are retained. The history-training comparison uses one shared initialization. Calibration scores and retained intervals are descriptive, not selection-independent evidence of superiority. The score has not been validated against human preference or executed trajectories. The motion-quality cohort is separate, and the illustrated sequence is a selected qualitative case. Its complete source and selection record are retained with the evidence.

VI. CONCLUSION

ForeDance brings anticipatory music conditioning to structured streaming dance: a frozen music foundation model predicts prospective context, kinematics-guided structure–detail training organizes motion components, and Commit Forcing learns continuation after their joint commitment. Completed studies find consistent soundtrack correspondence and characterize motion quality under matched history-training choices. Together, the method and reference interface provide an explicit framework for studying how musical foresight translates into continuously generated robot motion.

APPENDIX I

REPRESENTATION AND TRAINING DETAILS

a) *Native representation and boundary state:* We represent motion directly in the robot’s configuration space, including full pelvis orientation. A fixed yaw anchor R_A is removed once for a motion sequence, and every frame stores absolute orientation in that anchor:

$$R_t^A = R_A^\top R_t^{\text{world}}, \quad (9)$$

$$x_t = [\Delta p_{xy,t}^{\text{local}}, h_t, \rho(R_t^A), \theta_t]. \quad (10)$$

Here $\Delta p_{xy,t}^{\text{local}}$ denotes planar displacement, h_t is pelvis height, ρ is the six-dimensional rotation representation [43], and θ_t contains 29 joint angles, giving 38 motion dimensions. This retains roll and pitch without multiplying a sequence of predicted rotation increments. Planar displacements are integrated using the appropriate preceding heading. The anchor remains fixed over the generated sequence; it is not reset at every commit.

The corresponding 71-dimensional boundary state is

$$b_t = [h_t, v_{xy,t}^{\text{local}}, v_{z,t}, \rho(R_t^A), \omega_t, \theta_t, \dot{\theta}_t]. \quad (11)$$

The state contains pelvis height, local planar and vertical velocities, orientation, angular velocity ω_t , joint angles, and

joint velocities. Conditioning on these quantities makes the starting pose and motion explicit at each segment boundary. They are reconstructed from the committed reference.

b) Streaming configuration: The motion rate is 30 Hz, with two frames per latent step. We use 64 history steps (128 frames, approximately 4.27 s), an eight-step plan (16 frames, approximately 0.53 s), and a four-step commit (eight frames, approximately 0.27 s). The codec latent has 16 dimensions and its codebook has 512 entries. The music condition uses 14 native forecast states at 25 Hz, spanning approximately 0.56 s. Each state retains its physical timestamp.

c) Codec objective: Equation (5) describes the forward reconstructions. For training, let sg denote stop-gradient and set $r = z - \text{sg}(e(q))$. The actual decoder inputs are

$$z_f = \text{sg}(e(q)) + r, \quad (12)$$

$$z_s = z + \text{sg}(e(q) - z). \quad (13)$$

The full path equals z and sends its encoder gradient through r . The structural path equals the codebook embedding in the forward pass and uses a straight-through encoder gradient. Both paths train the same decoder. An auxiliary path perturbs the residual before the same decoder:

$$x_c = D_\psi(\text{sg}(e(q)) + \mathcal{C}(r), b). \quad (14)$$

Let Φ_f contain all coordinates of the four kinematic views, and let Φ_s retain their structural coordinates. The full reconstruction is supervised on Φ_f ; the structure-only and corrupted reconstructions are supervised on Φ_s . The combined objective is

$$\mathcal{L}_{\text{codec}} = \mathcal{L}_f + \mathcal{L}_s + 0.25\mathcal{L}_{\text{robust}} + \mathcal{L}_{\text{VQ}} + \mathcal{L}_{\text{boundary}}. \quad (15)$$

The following paragraphs specify each term and the coordinate partitions.

d) Conditioning and sampling: Figure 2 and Eqs. (2)–(3) define timestamp alignment and residual modulation. Motion queries occur at $d_i = i/15$ seconds for $i = 1, \dots, 8$; interpolation uses the forecast states’ actual lead times. A null condition bypasses the modulation exactly. The structural planner uses a classification objective and the detail model a denoising objective. Residual generation uses ten DDIM steps [44].

e) Kinematic reconstruction losses: The full view contains four blocks: native motion, native velocity, forward-kinematics positions, and geometry velocity. Each block is standardized with training-split statistics. For a view Φ with blocks $\Phi^{(k)}$, define the reconstruction error

$$\mathcal{E}_\Phi(\hat{x}, x) = \frac{1}{4} \sum_{k=1}^4 \text{mean} \left[\left(\Phi^{(k)}(\hat{x}) - \Phi^{(k)}(x) \right)^2 \right]. \quad (16)$$

This gives $\mathcal{L}_f = \mathcal{E}_{\Phi_f}(x_f, x)$, $\mathcal{L}_s = \mathcal{E}_{\Phi_s}(x_s, x)$, and $\mathcal{L}_{\text{robust}} = \mathcal{E}_{\Phi_s}(x_c, x)$. Equal block weights prevent the number of geometric coordinates from determining their relative contribution. The VQ term is the codebook loss plus 0.25 times the commitment loss.

For the auxiliary robustness objective, the residual is zeroed with probability 0.5. Otherwise, Gaussian noise is added with scale sampled uniformly from 0.01–0.05 times the residual channel standard deviation. The resulting reconstruction is supervised on the structural view. This codec objective is separate from FHC, which regularizes the generator’s motion history.

The boundary loss supervises the first three decoded frames relative to the supplied boundary. It combines normalized smooth- L_1 errors for velocity, acceleration, and jerk with weights 1, 0.5, and 0.25, respectively, plus pelvis-rotation geodesic error with weight 1. These quantities explicitly penalize a reconstructed segment whose beginning is inconsistent with the target transition.

f) Body-part supervision views: In the native block, structure selects the first 24 of 38 coordinates: nine root coordinates, twelve leg joints, and three waist joints. The remaining 14 coordinates are the bilateral shoulder (three per arm), elbow (one), and wrist (three) joints. In the 35-dimensional velocity block, structure selects the first 21 coordinates: root linear/angular velocity and the same 15 leg/waist joint velocities. The other 14 are arm-joint velocities.

The geometry block contains 31 points: 29 non-root body origins and two lowest-foot geometry points. Structure selects the torso, the two wrist-yaw body origins, and the two lowest-foot points. Detail selects the remaining 26 points. Geometry velocity uses the same partition. These are supervision and analysis views, not independent anatomical decoders. In particular, the structural geometry view can constrain wrist placement while the complementary native view describes arm articulation.

g) Generator optimization and initialization: Generator training uses 6,250 updates with batch size 48, corresponding to 300,000 training windows. Mixed-start training draws 20% song-origin starts and 80% continuation windows; continuation-start training uses ordinary windows throughout. Residual sampling uses guidance fixed at 1.0; the music model remains frozen during generator training.

h) History regularization: FHC applies to the complete 64-step history with probability 0.5 per example. The discrete embeddings and continuous residual history share a severity timestep sampled from 850–999, with independent Gaussian noise. Their masks and timestamps are preserved, as are the boundary state and forecast conditions. Corruption is applied in the supervised pass after CoF context construction and is disabled at inference.

i) Constructing CoF transitions: Figure 4 combines context construction and target rebasing. The context-replacement decision selects the entire history–state pair together. Both the first-pass commitment and the sampled next structure are obtained without gradients. Structural cross-entropy retains the recorded target IDs; residual denoising uses the target of Eq. (7) under the same sampled structure. The recorded target is not re-encoded with the generated boundary state. FHC is applied after this context selection, keeping target construction separate from history

regularization.

APPENDIX II EVALUATION PROTOCOL

The reported correspondence result uses 99 trajectories: three generator training seeds, three sampling seeds, and eleven tracks. Each trajectory contains 1,800 frames at 30 Hz and supplies 56 overlapping five-second windows with one-second stride. For every motion, the intended audio is compared with all ten wrong-track donors at matched relative windows, preserving each source’s native audio offset. Means are formed over windows and donors, then sampling repeats within training-seed/query-track cells, and finally the three training seeds and 11 query tracks. Each bootstrap draw resamples three training seeds and eleven query tracks with replacement; the same sampled query-track list is applied across the selected training seeds. Percentiles 2.5 and 97.5 over 10,000 draws define descriptive 95% intervals. The gallery of donors remains fixed. The separate 18-track motion-quality cohort uses common source-supported 23 s segments beginning at frame 264, three training seeds and three sampling seeds, yielding 162 outputs per configuration. Its scores and uncertainty are kept separate from the 60 s correspondence study.

A. Music–Motion Evaluator

a) *Frozen representation and readout*: The G1-native evaluator receives paired five-second windows: 150 audio feature frames at 30 Hz and 100 motion feature frames at 20 Hz. Motion features are extracted once from each complete trajectory, then normalized with frozen training-split statistics before windowing. The continuous 357-dimensional frontend retains full root rotation in $SO(3)$; it does not reset the orientation anchor separately for every scoring window. The two encoders produce deterministic posterior means for each window in FP32 inference. For evaluator instance k , denote these embeddings by $\mu_w^{a,k}$ and $\mu_w^{x,k}$. The adopted score is

$$s_w(x, a) = \frac{1}{3} \sum_{k=1}^3 \frac{\langle \mu_w^{x,k}, \mu_w^{a,k} \rangle}{\|\mu_w^{x,k}\|_2 \|\mu_w^{a,k}\|_2}. \quad (17)$$

Scores are averaged across the three independently trained evaluator instances before pairwise ranking; embeddings and final accuracies are not averaged instead. The checkpoints use standard content-group sampling and the final training update (14,500) for each seed, with no best-seed choice. Encoder weights and normalization remain fixed throughout generator scoring. An absolute score difference of at most 10^{-7} is treated as a tie and receives half credit. This specifies the correspondence statistic in Eq. (8) without conflating it with retrieval recall.

b) *Evaluator validation data*: Section IV-B reports the evaluator’s reference-motion results on 30 content groups and 3,661 windows. Each discrimination task uses 15 disjoint song pairs. These data are separate from the generated-motion gallery. The adopted score-level ensemble and all candidate comparisons are retained in the evidence record;

calibration does not establish perceptual validity on executed robot trajectories.

c) *Reproducible score aggregation*: The reported generator configuration supplies nine training/sampling combinations, each with an $11 \times 11 \times 56$ music–motion score array. The diagonal contains intended pairs; each off-diagonal column supplies the ten alternative tracks for one fixed motion. All 99 per-motion scores, margins, and accuracies are reproduced from these arrays. No source or window is removed based on its score. The complete two-configuration archive and metric-specific reporting choices remain in the evidence record.

d) *G1 distribution and local-motion features*: Kinetic descriptors summarize motion dynamics, while geometric descriptors summarize body configurations and spatial relations in the G1 domain. FID compares feature means and covariances with those of retargeted references; diversity measures pairwise distances in the same feature space. Extractor version, coordinate convention, normalization, crop length, and sample counts are part of each result. The earlier three-model table and current ForeDance results are not pooled: their numerical feature-scale equivalence has not been established. A multi-seed FID aggregation must retain how collections were formed; it is not a mean of single-song FIDs. For future execution scoring, generation and execution must use identical feature extraction. Exact feature lists and activity/contact thresholds require the corresponding implementation receipt.

e) *Beat alignment*: For motion-event times $\mathcal{B}_x$ and music-beat times $\mathcal{B}_a$, the motion-to-music BAS definition is

$$\text{BAS} = \frac{1}{|\mathcal{B}_x|} \sum_{b \in \mathcal{B}_x} \exp\left(-\frac{\min_{a \in \mathcal{B}_a} (b - a)^2}{2\sigma^2}\right). \quad (18)$$

The displayed BAS is the archived forward-kinematics motion-to-music score. At 30 Hz, mean keypoint speed is smoothed with a Gaussian of standard deviation five frames; strict local minima define motion beats. The matching kernel uses $\sigma = 3/30 = 0.10$ s. Music beats retain the archived audio-beat extraction and source offset. Empty music- or motion-beat sets receive zero. Scores are averaged over tracks and sampling repeats within each training seed, then across the three training seeds. The separately archived music-to-motion BAS uses the reverse matching direction and is not substituted into this column. Event counts, empty-event handling, smoothing, and any beat-matching tolerance must accompany reproducible scores. Beat precision/recall are auxiliary coverage diagnostics, not requirements that dance hit every audio beat. The layer analyzer’s impact proxy is a smoothed absolute derivative of the speed curve, not contact force. Zero-lag correlation with audio onset strength measures aligned dynamic response; best-lag correlation and its lag are diagnostics, not permission to retune primary music scores.

TABLE IV

FULL SOUNDTRACK-CORRESPONDENCE STATISTICS. CORRECT AND WRONG DENOTE COSINE SCORES WITH INTENDED AND ALTERNATIVE GALLERY MUSIC. MARGIN IS CORRECT MINUS WRONG. PAIRWISE ACCURACY GIVES HALF CREDIT TO TIES; INTERVALS ARE DESCRIPTIVE 95% BOOTSTRAP INTERVALS WITH THE GALLERY FIXED.

Method	Correct	Wrong	Margin	Margin interval	Pairwise accuracy (%) [interval]
ForeDance	0.2057	0.0274	+0.1783	[0.0983, 0.2608]	75.01 [65.17, 84.13]

APPENDIX III DETAILED CORRESPONDENCE STATISTICS

Table IV retains the component scores and margin intervals underlying Table II. Correct-minus-wrong means are positive for every generator training seed. These are the same 99 trajectories and the same fixed-gallery aggregation described in Appendix II. The companion scores and intervals use the same configuration as the displayed correspondence accuracy; they are not maximized independently. Intervals are descriptive conditional summaries of that configuration.

APPENDIX IV REFERENCE DELIVERY AND SUPPORTING EXECUTION DIAGNOSTICS

a) Replay setup: Four generated trajectories from one FineDance track and one sampling seed are replayed in MuJoCo for approximately 60 s each.

b) Reference construction and delivery: A four-step latent commit contains eight 30 Hz frames. A continuous 50 Hz time grid yields 13 or 14 reference samples per commit without resetting the sampling phase at each packet. The controller receives committed references rather than motor torques. The four diagnostic replays use the same fixed SONIC controller at real-time playback, with no uncommitted-plan preview, clipping, startup ramp, or alignment hold. Initial state, delivered future-reference access, and packet timestamps are part of the protocol; full-trajectory delivery and incremental committed-motion delivery are not interchangeable controls. The trajectories begin at an artificial continuation point with empty motion history, rather than at full-song onset.

c) Observed layers: Let M_{gen} , M_{sent} , M_{target} , and M_{exec} denote generated, resampled, consumed, and measured motion. Target and measured joints use named fields in MuJoCo order, with matched coordinate conventions, offsets, and timestamps.

d) Conditions and measurements: The first three trajectories share cached forecast content but use different age traces: ideal age, the earlier measured trace, and a revised measured trace. The fourth uses live forecast content with the revised trace. These labels identify the generation conditions of the saved trajectories.

For active joint j , amplitude and velocity-variation reten-

tion are

$$R_j^{\text{amp}} = \frac{P_{95}(q_j^{\text{exec}}) - P_5(q_j^{\text{exec}})}{P_{95}(q_j^{\text{target}}) - P_5(q_j^{\text{target}})}, \quad (19)$$

$$R_j^{\text{dyn}} = \frac{\mathbb{E}[(\dot{q}_j^{\text{exec}})^2]}{\mathbb{E}[(\dot{q}_j^{\text{target}})^2]}. \quad (20)$$

Only joints with a target percentile range above 0.02 rad enter the active-joint summary, and ratios with near-zero denominators are excluded. The reported retention values are medians of joint-wise ratios. The dynamic ratio measures squared joint velocity; it is not physical energy or a whole-body energy ratio. Joint-position RMSE is evaluated after time alignment. Alignment lag is estimated within ± 0.5 s on the 50 Hz grid; the four recorded estimates are 40 ms, which is a case-specific alignment result rather than a general controller or system latency. Steady-state summaries exclude the first two seconds, so they do not establish startup behavior.

e) Interpretation: Amplitude retention ranges from 0.858 to 0.901, while squared-velocity retention ranges from 0.411 to 0.496. Position tracking thus coexists with dynamic attenuation. High-frequency variation can include articulation and jitter; these measurements alone do not determine the musical value of the attenuation.

TABLE V

SUPPORTING FIXED-REFERENCE REPLAY MEASUREMENTS. DATA FROM THE EARLIER 34D GENERATOR, SEPARATE FROM THE CURRENT 38D COHORT. EACH ROW IS ONE MUJoCo REPLAY THROUGH FIXED SONIC, EXCLUDING ITS FIRST TWO SECONDS; MUSIC INFERENCE IS NOT RUN DURING REPLAY.

Reference-generation condition	Aligned RMSE (rad)	Median R^{amp}	Median R^{dyn}
Cached forecast, ideal age	0.1421	0.869	0.496
Cached forecast, earlier age trace	0.1281	0.858	0.428
Cached forecast, revised age trace	0.1394	0.877	0.412
Live forecast, revised age trace	0.1431	0.901	0.411

REFERENCES

- [1] R. Li, S. Yang, D. A. Ross, *et al.*, “AI Choreographer: Music conditioned 3D dance generation with AIST++,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 13 401–13 412.
- [2] J. Tseng, R. Castellon, and C. K. Liu, “EDGE: Editable dance generation from music,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 448–458.
- [3] R. Li, J. Zhao, Y. Zhang, *et al.*, “FineDance: A fine-grained choreography dataset for 3D full body dance generation,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 10 234–10 243.
- [4] X. Li, R. Li, S. Fang, *et al.*, “Music-aligned holistic 3D dance generation via hierarchical motion modeling,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025, pp. 14 420–14 430. [Online]. Available: https://openaccess.thecvf.com/content/ICCV2025/html/Li_Music-Aligned_Holistic_3D_Dance_Generation_via_Hierarchical_Motion_Modeling_ICCV_2025_paper.html
- [5] K. Ji, B. Qian, B. Wu, *et al.*, “DiscoForcing: A unified framework for real-time audio-driven character control with diffusion forcing,” *arXiv preprint arXiv:2605.28491*, 2026. [Online]. Available: <https://arxiv.org/abs/2605.28491>
- [6] Magenta Team, “Magenta RealTime 2: Open & local live music models,” Magenta blog, June 2026. [Online]. Available: <https://magenta.withgoogle.com/magenta-realtime-2>
- [7] H. Wang, X. Chen, J. Zhang, *et al.*, “DC-Motion: Decoupling structure and details via discrete-continuous tokens for human motion generation,” *arXiv preprint arXiv:2606.14721*, 2026. [Online]. Available: <https://arxiv.org/abs/2606.14721>
- [8] Z. Luo, Y. Yuan, T. Wang, *et al.*, “SONIC: Supersizing motion tracking for natural humanoid whole-body control,” *Science Robotics*, vol. 11, no. 117, p. eaed4592, 2026. [Online]. Available: <https://arxiv.org/abs/2511.07820>
- [9] B. McFee, C. Raffel, D. Liang, *et al.*, “librosa: Audio and music signal analysis in Python,” in *Proceedings of the 14th Python in Science Conference*, 2015, pp. 18–25. [Online]. Available: <https://proceedings.scipy.org/articles/Majora-7b98e3ed-003>
- [10] Y. Li, R. Yuan, G. Zhang, *et al.*, “MERT: Acoustic music understanding model with large-scale self-supervised training,” in *International Conference on Learning Representations*, 2024. [Online]. Available: https://proceedings.iclr.cc/paper_files/paper/2024/hash/33dffa2e3d2ab74a783d1a8c292f66d9-Abstract-Conference.html
- [11] H. Zhu, Y. Zhou, H. Chen, *et al.*, “MuQ: Self-supervised music representation learning with mel residual vector quantization,” *arXiv preprint arXiv:2501.01108*, 2025. [Online]. Available: <https://arxiv.org/abs/2501.01108>
- [12] L. Siyao, W. Yu, T. Gu, *et al.*, “Bailando: 3D dance generation by actor-critic GPT with choreographic memory,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 11 050–11 059.
- [13] R. Li, Y. Zhang, Y. Zhang, *et al.*, “LODGE: A coarse to fine diffusion network for long dance generation guided by the characteristic dance primitives,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 1524–1534.
- [14] K. Yang, X. Tang, Z. Peng, *et al.*, “MEGADance: Mixture-of-experts architecture for genre-aware 3D dance generation,” in *Advances in Neural Information Processing Systems*, vol. 38, 2025. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2025/hash/102bd3c1076dfe98043b19340ac44e08-Abstract-Conference.html
- [15] P. Dhariwal, H. Jun, C. Payne, *et al.*, “Jukebox: A generative model for music,” *arXiv preprint arXiv:2005.00341*, 2020. [Online]. Available: <https://arxiv.org/abs/2005.00341>
- [16] X. Liu, Z. Feng, D. Kanojia, *et al.*, “DGFM: Full body dance generation driven by music foundation models,” *arXiv preprint arXiv:2502.20176*, 2025. [Online]. Available: <https://arxiv.org/abs/2502.20176>
- [17] N. Nakahara, K. Miyazaki, H. Sakamoto, *et al.*, “Dance motion control of a humanoid robot based on real-time tempo tracking from musical audio signals,” in *Entertainment Computing – ICEC 2009*, ser. Lecture Notes in Computer Science, vol. 5709, 2009, pp. 36–47. [Online]. Available: https://ist.ksc.kwansei.ac.jp/~nagata/data/2009_nakahara_ICEC2009.pdf
- [18] Lyria Team, A. Caillon, B. McWilliams, *et al.*, “Live music models,” *arXiv preprint arXiv:2508.04651*, 2025. [Online]. Available: <https://arxiv.org/abs/2508.04651>
- [19] A. van den Oord, O. Vinyals, and K. Kavukcuoglu, “Neural discrete representation learning,” in *Advances in Neural Information Processing Systems*, vol. 30, 2017. [Online]. Available: <https://proceedings.neurips.cc/paper/2017/hash/7a98af17e63a0ac09ce2e96d03992fbc-Abstract.html>
- [20] J. Zhang, Y. Zhang, X. Cun, *et al.*, “Generating human motion from textual descriptions with discrete representations,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 14 730–14 740.
- [21] C. Guo, Y. Mu, M. G. Javed, *et al.*, “MoMask: Generative masked modeling of 3D human motions,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 1900–1910.
- [22] X. Chen, B. Jiang, W. Liu, *et al.*, “Executing your commands via motion diffusion in latent space,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 18 000–18 010.
- [23] J. Cho, J. Kim, J. Kim, *et al.*, “DisCoRD: Discrete tokens to continuous motion via rectified flow decoding,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025, pp. 14 602–14 612.
- [24] Q. Zhu, H. Zhang, M. Lan, *et al.*, “Neural categorical priors for physics-based character control,” *ACM Transactions on Graphics*, vol. 42, no. 6, 2023.
- [25] Y. Shi, Y. Jiang, C. Tessler, *et al.*, “GPC: Large-scale generative pretraining for transferable motor control,” in *ACM SIGGRAPH 2026 Conference Papers*, 2026.
- [26] S. Bengio, O. Vinyals, N. Jaitly, *et al.*, “Scheduled sampling for sequence prediction with recurrent neural networks,” in *Advances in Neural Information Processing Systems*, vol. 28, 2015, pp. 1171–1179.
- [27] A. Lamb, A. Goyal, Y. Zhang, *et al.*, “Professor forcing: A new algorithm for training recurrent networks,” in *Advances in Neural Information Processing Systems*, vol. 29, 2016, pp. 4601–4609.
- [28] B. Chen, D. Martí Monsó, Y. Du, *et al.*, “Diffusion forcing: Next-token prediction meets full-sequence diffusion,” in *Advances in Neural Information Processing Systems*, vol. 37, 2024, pp. 24 081–24 125.
- [29] B. Jiang, X. Chen, A. Zeng, *et al.*, “Causal motion tokenizer for streaming motion generation,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops*, 2025, pp. 2045–2055. [Online]. Available: https://openaccess.thecvf.com/content/ICCV2025W/I-HFM/html/Jiang_Causal_Motion_Tokenizer_for_Streaming_Motion_Generation_ICCVW_2025_paper.html
- [30] L. Xiao, S. Lu, H. Pi, *et al.*, “MotionStreamer: Streaming motion generation via diffusion-based autoregressive model in causal latent

- space,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025, pp. 10 086–10 096.
- [31] C. Chi, S. Feng, Y. Du, *et al.*, “Diffusion policy: Visuomotor policy learning via action diffusion,” in *Robotics: Science and Systems XIX*, 2023. [Online]. Available: <https://roboticsproceedings.org/rss19/p026.pdf>
 - [32] K. Black, M. Y. Galliker, and S. Levine, “Real-time execution of action chunking flow policies,” in *Advances in Neural Information Processing Systems*, vol. 38, 2025. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2025/hash/300ccb2187dedd4edcc07f7e76d8e553-Abstract-Conference.html
 - [33] Z. Luo, J. Cao, A. Winkler, *et al.*, “Perpetual humanoid control for real-time simulated avatars,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 10 895–10 904. [Online]. Available: https://openaccess.thecvf.com/content/ICCV2023/papers/Luo_Perpetual_Humanoid_Control_for_Real-time_Simulated_Avatars_ICCV_2023_paper.pdf
 - [34] M. Ji, X. Peng, F. Liu, *et al.*, “ExBody2: Advanced expressive humanoid whole-body control,” *arXiv preprint arXiv:2412.13196*, 2024. [Online]. Available: <https://arxiv.org/abs/2412.13196>
 - [35] Q. Liao, T. E. Truong, X. Huang, *et al.*, “BeyondMimic: From motion tracking to versatile humanoid control via guided diffusion,” *arXiv preprint arXiv:2508.08241*, 2025. [Online]. Available: <https://arxiv.org/abs/2508.08241>
 - [36] Z. Jiang, Y. Xie, J. Li, *et al.*, “Harmon: Whole-body motion generation of humanoid robots from language descriptions,” in *Proceedings of the 8th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 270. PMLR, 2025, pp. 3015–3026. [Online]. Available: <https://proceedings.mlr.press/v270/jiang25b.html>
 - [37] Z. Li, C. Chi, Y. Wei, *et al.*, “Do you have freestyle? expressive humanoid locomotion via audio control,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026, pp. 956–965. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2026/papers/Li_Do_You_Have_Freestyle_Expressive_Humanoid_Locomotion_via_Audio_Control_CVPR_2026_paper.pdf
 - [38] Y. Yuan, J. Song, U. Iqbal, *et al.*, “PhysDiff: Physics-guided human motion diffusion model,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 16 010–16 021. [Online]. Available: https://openaccess.thecvf.com/content/ICCV2023/html/Yuan_PhysDiff_Physics-Guided_Human_Motion_Diffusion_Model_ICCV_2023_paper.html
 - [39] Z. Zhang, K. Wen, M. Xu, *et al.*, “Learning whole-body humanoid locomotion via motion generation and motion tracking,” *arXiv preprint arXiv:2604.17335*, 2026. [Online]. Available: <https://arxiv.org/abs/2604.17335>
 - [40] M. Xu, Y. Shi, K. Yin, *et al.*, “PARC: Physics-based augmentation with reinforcement learning for character controllers,” in *ACM SIGGRAPH 2025 Conference Papers*, 2025.
 - [41] E. Perez, F. Strub, H. de Vries, *et al.*, “FiLM: Visual reasoning with a general conditioning layer,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 32, 2018, pp. 3942–3951. [Online]. Available: <https://ojs.aaai.org/index.php/AAAI/article/view/11671>
 - [42] J. Ho, A. Jain, and P. Abbeel, “Denoising diffusion probabilistic models,” in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 6840–6851. [Online]. Available: <https://proceedings.neurips.cc/paper/2020/hash/4c5bcfec8584af0d967f1ab10179ca4b-Abstract.html>
 - [43] Y. Zhou, C. Barnes, J. Lu, *et al.*, “On the continuity of rotation representations in neural networks,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019, pp. 5745–5753. [Online]. Available: https://openaccess.thecvf.com/content_CVPR_2019/papers/Zhou_On_the_Continuity_of_Rotation_Representations_in_Neural_Networks_CVPR_2019_paper.pdf
 - [44] J. Song, C. Meng, and S. Ermon, “Denoising diffusion implicit models,” in *International Conference on Learning Representations*, 2021.